

TECHNOLOGY EXPLORER SERIES

BOOK 3

ROBOT WORLD

Meet the Machines That Help Us



Fansci Solutions

Robot World: Meet the Machines That Help Us

The Secret Life of Technology, Book 3

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Series: The Secret Life of Technology

Book 1: The Secret Life of the Internet

Book 2: Inside a Smartphone: The Tiny World in Your Hand

Book 3: Robot World: Meet the Machines That Help Us

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Meet Your Guides

Welcome back, explorer! You have met the parts inside a smartphone and travelled through the Internet. Now it is time to step into a world of amazing machines that can move, think, and help us every day.



Dot	Returning guide — a glowing packet of information, always curious.
Bolt	A friendly factory robot arm — strong, precise, and loves building things.
Whirr	A buzzing drone — sees the world from above.
Sprocket	A wheeled rover — explores where humans cannot go.
Cora	A humanoid companion robot — learns from people and lends a hand.
Sparks	A micro-bot surgeon — tiny but incredibly careful.
Tread	A heavy-duty rescue robot — brave and unstoppable.
Echo	A voice-controlled home assistant robot — listens and learns.

Welcome

In our last adventures we explored the Internet and shrunk inside a smartphone. Now it is time to look at the machines that share our world — robots! They build our cars, deliver packages, explore the deep sea, and even help doctors in surgery. Ready? Follow Dot into Robot World!

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Chapter 1

What Is a Robot?

They build cars, explore volcanoes, vacuum floors, and even perform surgery. But what exactly is a robot?

Mission Start: Enter Robot World!

A New Adventure Begins

Sam was visiting the local science museum. In the main hall, a robotic arm was picking up coloured blocks and stacking them into a perfect tower. Beside it, a small wheeled robot was drawing a picture on the floor with a marker.

Sam stared, amazed:

“How do they know what to do? Nobody is holding a controller!”

A familiar glow appeared on the screen of a nearby display.

Dot

“Remember me? We have explored the Internet and the inside of a smartphone together. Now I want to show you something even more exciting — the world of robots!”

Three Things Every Robot Does

Not every machine is a robot. A toaster heats bread, but it is not a robot. So what makes something a robot? Every true robot does three things:

1. **Sense** — it collects information about the world using **sensors**.
2. **Think** — it processes that information using a **controller** (a kind of computer brain).
3. **Act** — it does something in the real world using **actuators** (motors, wheels, grippers).

Big Idea

A robot is a machine that can **sense** the world, **think** about what to do, and then **act** on its own.

Imagine This

Imagine you are playing football. Your eyes *sense* the ball coming. Your brain *thinks*: “I should kick it left.” Your leg *acts* and kicks. A robot works the same way — eyes, brain, body — just made of metal and code instead of flesh and bone.

Robot or Just a Machine?

Let us play a quick sorting game.

Machine	Robot?
A light switch	No
A self-driving car	Yes
A washing machine	No (it follows a fixed timer)
A robot vacuum that avoids furniture	Yes
A vending machine	No
A drone that follows you while filming	Yes

The key difference? Robots *respond* to what is happening around them. They make decisions.

Did You Know?

The word “robot” comes from the Czech word *robota*, meaning forced labour. It was first used in a 1920 play called *R.U.R.* by Karel Čapek.

Robots Come in All Shapes

Robots do not all look like metal people. In fact, most robots look nothing like humans at all!

- **Robot arms** — bolted to factory floors, they weld, paint, and assemble.
- **Wheeled robots** — rovers, delivery bots, robot vacuums.
- **Drones** — flying robots used for filming, deliveries, and search-and-rescue.
- **Humanoids** — shaped like people, designed to work alongside us.
- **Soft robots** — made of flexible materials, inspired by octopuses and worms.
- **Micro-bots** — tiny robots that can travel inside the human body.

Bolt

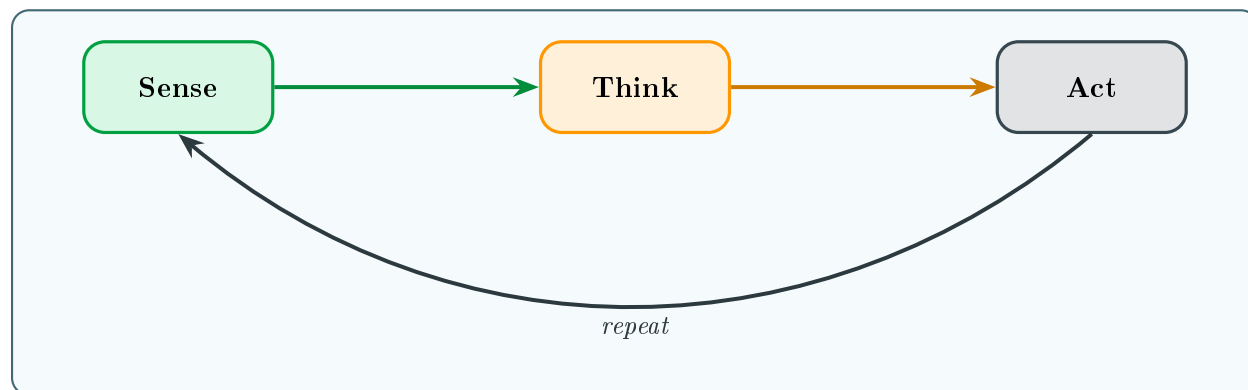
“I am a robot arm! I can lift heavy car doors all day long without getting tired. But I would be terrible at climbing stairs — that is a job for a different kind of robot.”

★ Did You Know?

There are over 3.9 million industrial robots working in factories around the world — that is roughly one robot for every 2,000 people on Earth. And the number is growing every year!

👍 Try It

Look around your home. Can you find anything that senses, thinks, and acts? A robot vacuum? A smart speaker? Make a list and decide: is it a true robot, or just a clever machine?



The sense–think–act loop that every robot follows.

📋 Quick Check

1. What three things does every robot do?
2. Name one job that is better for a robot than a human. Why?
3. Can you find one robot (or robot-like machine) in your home?



New Words

Robot	A machine that can sense, think, and act on its own.
Sensor	A part that detects things (light, sound, distance, touch).
Controller	The computer brain that makes decisions.
Actuator	A part that moves or does work (motor, gripper, propeller).
Humanoid	A robot shaped like a human body.
Drone	An unmanned flying robot.

Chapter 2

Robot Bodies — Arms, Wheels, and Legs

Humans have arms, legs, and hands. Robots have joints, wheels, and grippers. Let us explore how robot bodies are built to move.

Meet Bolt — the factory arm!

Built for a Job

A robot's body is designed for its task. A factory robot arm does not need legs — it stays in one place and reaches. A delivery robot does not need arms — it just needs wheels and a cargo box.

Bolt

“My body has six joints, just like your shoulder, elbow, and wrist — but doubled! That means I can twist and reach in ways no human arm can.”



Imagine This

Think of a crane at a building site. It is tall and has a long arm that swings. A robot arm is like a tiny, super-precise crane that sits on a table and can move to the exact same spot a thousand times without missing.

How Robots Move

Every moving robot uses some combination of these:

- **Motors** — electric motors spin wheels or rotate joints.
- **Joints** — hinges that let parts bend or twist.
- **Gears** — toothed wheels that change speed and force.
- **Wheels** — great for flat surfaces; fast and efficient.

- **Legs** — better for rough terrain but harder to control.
- **Tracks** — like tank treads; good on mud, rubble, or sand.
- **Propellers** — spinning blades that push air down to make drones fly.

★ Did You Know?

Boston Dynamics' robot dog "Spot" has 12 motors — three in each leg — and can climb stairs, open doors, and even dance!

Robot Hands and Grippers

Robots need to grab things too. But they do not all use fingers.

1. **Pincer grippers** — two fingers that squeeze together (like tongs).
2. **Suction cups** — use a vacuum to stick to flat surfaces (like glass).
3. **Magnetic grippers** — attract metal objects.
4. **Soft grippers** — inflate like balloons to wrap gently around odd shapes.
5. **Multi-finger hands** — five fingers controlled independently, the closest to a human hand.

💡 Big Idea

Engineers choose the simplest gripper that can do the job. More fingers means more complexity — and more things that can go wrong.

What Are Robots Made Of?

- **Aluminium and steel** — strong but light for arms and frames.
- **Plastic and carbon fibre** — lightweight for drones.
- **Silicone and rubber** — flexible, used in soft robots.
- **Titanium** — extremely tough, used in space robots.

Sprocket

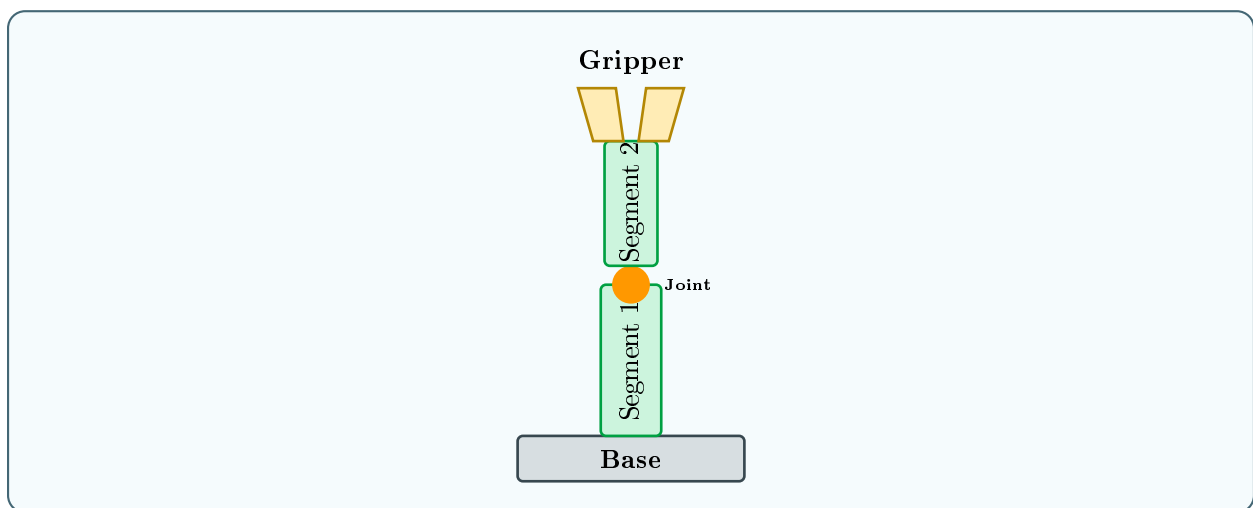
"I am made mostly of aluminium. That keeps me light enough to roll over sand dunes without sinking!"

Experiment

Build a simple gripper.

1. Take two wooden lolly sticks and a rubber band.
2. Place a small sponge between the tips.
3. Wrap the rubber band around the other end to hold them together.
4. Squeeze the sticks apart — the sponge tips close like a pincer.
5. Try picking up a cotton ball, a marble, and a pencil. Which is hardest?

This is how a simple pincer gripper works!



Parts of a typical robot arm.

What Powers a Robot?

Every motor, sensor, and controller needs **energy**. Most robots get their energy from **batteries** — portable packs of stored electricity.

- **Rechargeable batteries** — the most common; used in drones, vacuums, and rovers.
- **Mains power** — factory robots often plug into the wall for unlimited energy.
- **Solar panels** — some space robots collect energy from sunlight.
- **Charging docks** — home robots return to a base station when running low.



Imagine This

Imagine going to school without eating breakfast or lunch. By afternoon you would have no energy to run or think. A robot with a low battery is exactly the same — it slows down and eventually stops. Good robot designers plan carefully so their robot always has enough energy to finish the job *and* get back to its charger.



Did You Know?

A typical drone battery lasts only 20–30 minutes. That is why delivery drones must fly fast and land quickly!



Quick Check

1. What is the difference between a motor and a gear?
2. Name two types of gripper and when you might use each one.
3. Why do drones use propellers instead of wheels?



New Words

Motor	A device that converts electricity into spinning motion.
Joint	A connection between two parts that allows bending or rotating.
Gear	A toothed wheel that transfers force between shafts.
Gripper	A robot hand designed to grab objects.
Degrees of freedom	The number of independent ways a robot can move.
Chassis	The main frame or body of a robot.

Chapter 3

How Robots Sense the World

You have eyes to see, ears to hear, and skin to feel. Robots have sensors — electronic eyes, ears, and fingertips that help them understand what is happening around them.

Sensors: A Robot's Window to the World

Why Sensing Matters

Imagine trying to walk through your house with your eyes closed and your ears plugged. You would bump into everything! Robots without sensors would do the same.

Dot

“Sensors are the first step in the sense – think – act loop. Without them, a robot is just a lump of metal.”

Types of Robot Sensors

Cameras — Robot Eyes

Cameras capture pictures many times per second. A robot's computer analyses these images to recognise objects, read signs, or follow a line on the floor.

★ Did You Know?

Self-driving cars use up to eight **cameras** at once, giving them a 360-degree view — humans can only see about 180 degrees!

Everyday Device	Sensor Used	What It Detects
Automatic door	Infrared / motion	A person approaching
Smartphone	Accelerometer	Tilt (screen rotation)
Smoke alarm	Optical / ionisation	Smoke particles in air
Reversing car	Ultrasonic	Objects behind the car
Fitness tracker	Heart-rate sensor	Your pulse

Lidar — Laser Eyes

Lidar fires invisible **laser** beams in all directions and measures how long each beam takes to bounce back. This creates a **3D map** of the surroundings, even in the dark.

Ultrasonic Sensors — Echo Location

Just like a bat, these **ultrasonic** sensors send out a pulse of sound and listen for the echo. The time the echo takes to return tells the robot how far away an object is.

🔪 Imagine This

Stand in a large empty room and clap your hands. You hear the echo bounce off the walls. An ultrasonic sensor does the same thing, but millions of times faster and with sounds too high for your ears to hear.

Touch and Pressure Sensors

Some robots have sensitive skin. Pressure sensors tell them how hard they are gripping something — essential for a robot handing you a glass of water without crushing it!

Temperature Sensors

Useful for robots that cook, fight fires, or explore volcanoes. They measure heat without needing to touch the surface.

Gyroscopes and Accelerometers

These sensors tell a robot which way is up, how fast it is moving, and whether it is tilting. Drones would tumble out of the sky without them.

Whirr

“My gyroscope keeps me level, even when the wind blows. Without it I would flip upside down in seconds!”

Sensor Fusion — Combining Superpowers

One sensor alone can be fooled. A camera cannot see in the dark. Lidar cannot read colours. So engineers combine several sensors and let the computer brain mix all the data together. This is called **sensor fusion**.

💡 Big Idea

Sensor fusion means combining information from many sensors to build a more complete picture of the world — just as you use sight *and* hearing *and* touch at the same time.

🧪 Experiment

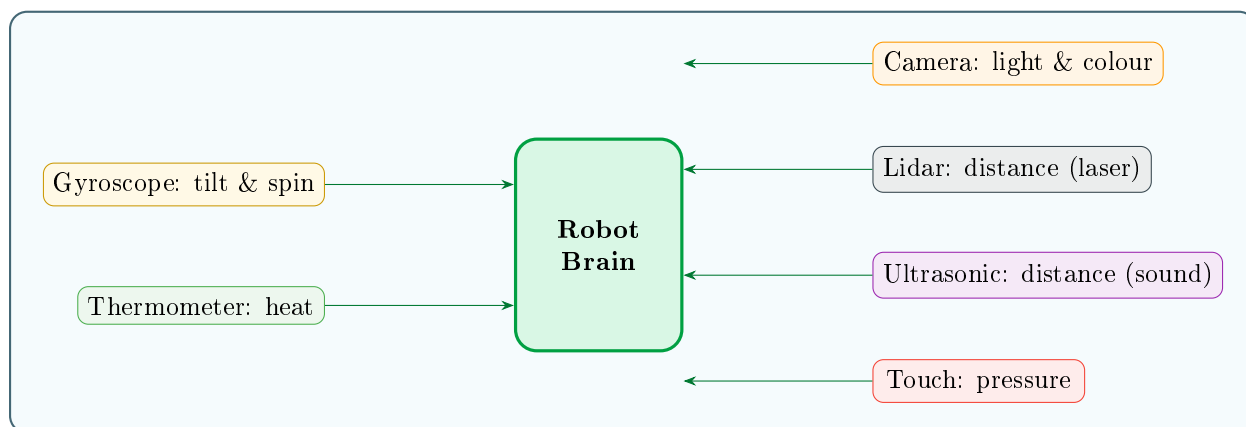
Test your own sensor fusion.

1. Close your eyes and try to touch your nose with one finger.
2. Now spin around three times, *then* try again.
3. Notice how much harder it is when your balance sensor (inner ear) is confused — even though your touch sensor still works.

Robots face the same challenge: if one sensor gives bad data, the brain must rely on the others.

📋 Quick Check

1. What does a lidar sensor measure, and how?
2. Why is sensor fusion better than using one sensor alone?
3. Which sensor would a robot need to detect a hot surface?



Common robot sensors and what they detect.



New Words

Sensor	A device that measures something in the environment.
Lidar	Light Detection and Ranging — uses lasers to map distances.
Ultrasonic	Using sound waves above human hearing to detect objects.
Sensor fusion	Combining data from multiple sensors for a better picture.
Gyroscope	A sensor that measures rotation and tilt.
Accelerometer	A sensor that measures changes in speed or direction.

Chapter 4

The Robot Brain

A robot can have the strongest arm and the sharpest sensors in the world, but without instructions it will just sit there doing nothing. The brain is what brings a robot to life.

Code: The Language Robots Understand

The Controller

Inside every robot is a small computer called the **controller**. It reads information from the sensors, decides what to do, and sends commands to the motors.



Imagine This

Imagine if your teacher gave the class no instructions for a science experiment. Everyone might mix different things at different times and the results would be chaos! A robot programme is like the experiment instructions that keep everything on track — a recipe the robot follows step by step.

Dot

“Remember the processor inside a smartphone? A robot’s controller is very similar — it follows instructions millions of times per second.”

How Do We Tell Robots What to Do?

Humans write instructions in a **programming language**. Popular ones for robots include:

- **Python** — easy to read, great for beginners.
- **C/C++** — fast, used in many industrial robots.

- **Scratch** — block-based, brilliant for learning logic.
- **ROS** — Robot Operating System, a toolkit engineers use to make different robot parts work together.

★ Did You Know?

The first programmable robot was “Unimate”, installed in a General Motors factory in 1961. It was programmed by manually moving its arm and recording the positions — no typing needed!

Algorithms — Step-by-Step Plans

An **algorithm** is a set of clear steps that solves a problem. Every robot programme is made of algorithms.

Quick Map

Algorithm: avoid an obstacle

1. Move forward.
2. If the distance sensor reads less than 30 cm, stop.
3. Turn right 90 degrees.
4. Go back to step 1.

Loops and Decisions

Two superpowers of every programme:

- **Loops** — repeat something over and over. Example: “Keep spinning the propellers until I say stop.”
- **Conditions** — choose what to do based on information. Example: “IF the floor sensor sees a cliff, THEN stop and turn around.”

Big Idea

Every clever thing a robot does is built from simple building blocks: sequences, loops, and conditions. Stack enough of them together and you get a machine that *looks* intelligent.

Try It

Write an algorithm for making a sandwich. Be very precise — remember, a robot cannot guess! For example:

1. Pick up bread with left gripper.
2. Place bread on plate.
3. Pick up butter knife...

How many steps did you need?

Experiment

Programme a human robot.

1. Ask a friend to be the “robot” and blindfold them (or close their eyes).
2. Place a target (a book or toy) across the room.
3. You are the programmer — you may only say: **Forward**, **Stop**, **Turn left**, or **Turn right**.
4. Try to guide your robot to touch the target.
5. Count how many commands it takes.

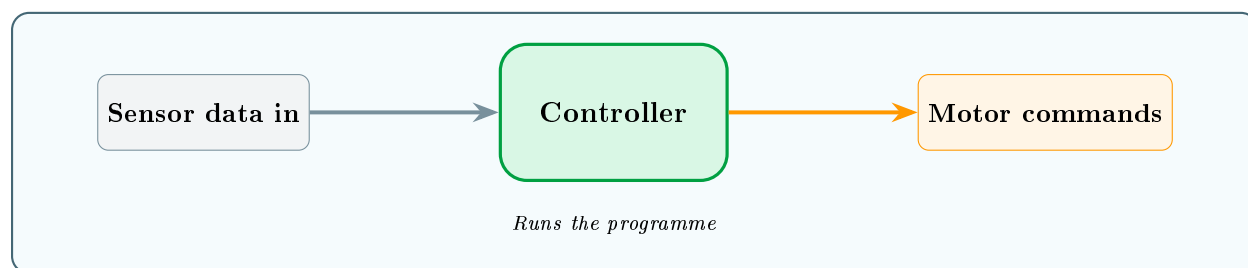
Notice how precise you must be — saying “go over there” means nothing to a robot! This is exactly why programmes need clear, step-by-step instructions.

Can Robot Brains Learn?

Most robots follow fixed programmes. But some newer robots can *learn* from examples — a bit like practising until you get better at a video game.

- A robot arm might try a thousand ways to pick up an egg until it finds the gentlest grip.
- A delivery robot might learn which routes have fewer obstacles.

This ability is called **machine learning**, and it is a type of **artificial intelligence** (AI). We will explore AI much more in Book 4: *When Machines Start Thinking!*



Inside a robot controller: sensors in, decisions made, commands out.



Quick Check

1. What is an algorithm?
2. Name the two programming superpowers mentioned in this chapter.
3. Why does a robot need very precise instructions?



New Words

Controller	The computer board inside a robot that runs the programme.
Programme	A set of instructions written in a programming language.
Algorithm	A step-by-step method for solving a problem.
Loop	An instruction that repeats until a condition is met.
Condition	A yes/no check that changes what the programme does next.
ROS	Robot Operating System — a popular framework for robot software.

Chapter 5

Factory Robots

Walk into a car factory and you will see hundreds of robot arms welding, painting, and bolting — all day, all night, never stopping for a snack.

Bolt's Home: The Assembly Line

The Robot Factory

Factories were the first places robots found real work. In the 1960s, engineers realised that robots could do repetitive, dangerous, or extremely precise jobs better than humans.

Bolt

"This is where I live! I weld 400 metal joints every hour. My human teammates handle the creative parts — I handle the heavy lifting."

Jobs on the Line

- **Welding** — joining metal pieces with intense heat.
- **Painting** — spraying perfectly even coats of paint.
- **Assembly** — fitting parts together (screws, clips, glue).
- **Pick and place** — grabbing items from a conveyor belt and placing them in boxes.
- **Quality inspection** — using cameras to spot defects smaller than a grain of sand.
- **Palletising** — stacking boxes onto pallets ready for shipping.

★ Did You Know?

A single car factory can have over 1,000 robots. Together they can build a complete car body in under one minute!

Cobots — Robots That Work With People

Older factory robots work behind safety cages because they are so strong they could hurt someone by accident. But a new generation of **collaborative robots** (cobots) are designed to work *beside* humans safely.

Cobots have:

- Padded, rounded bodies (no sharp edges).
- Force sensors that stop them instantly if they bump someone.
- Slower movements in shared workspaces.

💡 Big Idea

Cobots do not replace humans — they become teammates. The cobot handles heavy or boring tasks while the human does the thinking, checking, and problem-solving.

Warehouse Robots

When you order something online, a warehouse robot may help find it. Fleets of small orange robots at companies like Amazon carry entire shelving units to human packers, saving them thousands of steps a day.

Imagine This

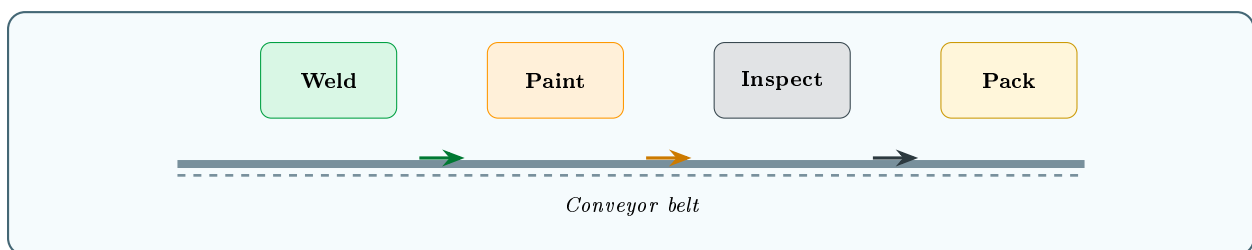
Imagine a library where the bookshelves come to *you* instead of you walking to the shelves. That is what warehouse robots do with products.

How an Assembly Line Works

Quick Map

A car on the assembly line:

1. Robot welder joins the metal body panels.
2. Robot arm bolts doors into place.
3. Painting robot sprays three coats of colour.
4. Camera robot inspects for scratches or dents.
5. Human worker adds seats, wiring, and trim.
6. Robot lifts the engine into position.
7. Quality robot runs a final check.
8. Finished car rolls off the line!



A simplified assembly line: products move past robot stations.

Imagine This

Imagine if every time you made a sandwich, you had to walk to a different room for bread, then another for butter, then another for cheese. An assembly line brings everything to you in order — that is why factories can build a car in under a minute.

Try It

Set up a mini assembly line at home.

1. Gather ten identical objects (LEGO bricks, cards, or coins).
2. Time yourself placing them into a box one by one.
3. Now ask a friend to pass them to you (like a cobot helping).
4. Is the teamwork version faster? By how much?

Quick Check

1. What is the difference between a robot and a cobot?
2. Name three jobs that factory robots do on an assembly line.
3. How do warehouse robots save workers thousands of steps?

New Words

Assembly line	A production system where products move past stations, each adding a part.
Welding	Joining two metal pieces by melting their edges together.
Cobot	A collaborative robot designed to work safely alongside humans.
Pick and place	A robot task: grab an item from one spot and put it in another.
Conveyor belt	A moving belt that carries items through a factory.
Palletising	Stacking finished goods onto a pallet for transport.

Chapter 6

Robots at Home

You might already live with a robot and not even realise it. From vacuum cleaners to lawn mowers, robots are quietly helping families every day.

Echo's Domain: The Smart Home

Robots Under Your Roof

Factory robots are big and expensive. Home robots are small, affordable, and designed to be safe around children and pets.

Echo

"I listen for your voice and control the smart lights, play music, and even remind you about homework. I am not just a speaker — I am a helper!"

Types of Home Robots

Robot Vacuums

These disc-shaped **robot vacuums** zip around your floors, sucking up dust and crumbs. They use **bump sensors**, **cliff sensors** (so they do not fall downstairs), and sometimes **lidar** to map your rooms.

★ Did You Know?

The first Roomba was sold in 2002. Since then, over 40 million robot vacuums have been sold worldwide.

Robot Lawn Mowers

Like a vacuum for your garden. They follow a boundary wire buried in the ground and mow a little bit every day, keeping the grass perfectly short without you lifting a finger.

Robot Window Cleaners

These cling to glass using suction and crawl across windows, spraying and wiping as they go — perfect for hard-to-reach places.

Smart Speakers and Assistants

Devices like Alexa, Google Home, and Siri are not physical robots, but they use the same sense–think–act pattern: microphone (sense), AI brain (think), speaker output and smart-home commands (act).

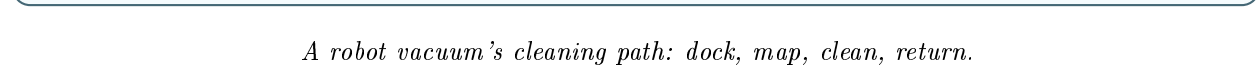
Companion Robots

Some robots are built purely to be friends. They chat, play games, and even recognise your face and mood.



Imagine This

Imagine a pet that never needs feeding, never makes a mess, and always wants to play. Companion robots are a bit like that — though real pets still give the best cuddles!



Quick Map

How a robot vacuum cleans a room:

1. Leaves its **charging dock**.
2. Spins its lidar sensor to build a map of the room.
3. Plans an efficient zig-zag path.
4. Drives forward, brushing and sucking dirt.
5. If it senses a chair leg or wall, it adjusts course.
6. When battery runs low, returns to the dock to recharge.
7. Resumes cleaning where it left off.

Experiment

Map your room like a robot vacuum.

1. Blindfold yourself (or close your eyes).
 2. Starting from the door, walk slowly with arms outstretched.
 3. Every time you touch something, say what it is and mark it on paper.
 4. After five minutes, open your eyes and compare your mental map to the real room.
- A robot vacuum builds its map the same way — by sensing obstacles and remembering where*

they are.

Staying Safe with Home Robots

- Keep small toys off the floor so vacuum robots do not choke on them.
- Never ride on a robot vacuum — it is not a hoverboard!
- Be careful with voice assistants: do not share personal information like passwords or your home address out loud.

Big Idea

Home robots save time on boring chores so you have more time for fun, family, and creativity. But they still need humans to set them up, maintain them, and make sure they behave.

Quick Check

1. How does a robot vacuum know where it has already cleaned?
2. Why must a robot vacuum return to its charging dock?
3. Name one safety rule for living with home robots.

New Words

Robot vacuum	An autonomous disc-shaped robot that cleans floors.
Boundary wire	A buried wire that tells a mowing robot where to stop.
Charging dock	A station where a robot returns to refill its battery.
Companion robot	A robot designed for social interaction and company.
Smart speaker	A voice-controlled assistant with microphone and speaker.
Autonomy	The ability to act without human control.

Chapter 7

Robots in Space and Deep Sea

Some places are too hot, too cold, too far, or too deep for humans. That is where robots really shine — they go where we cannot.

Sprocket and Tread: Exploring the Unknown

Robots in Space

Space is one of the most dangerous environments imaginable: no air, extreme temperatures, and deadly **radiation**. Sending a robot instead of a person is often safer, cheaper, and sometimes the only option.

Sprocket

*“I am inspired by the **Mars rovers**! They roll across another planet, drill into rocks, and send photos back to Earth — all by themselves, because radio signals take up to 20 minutes to travel between Mars and Earth.”*

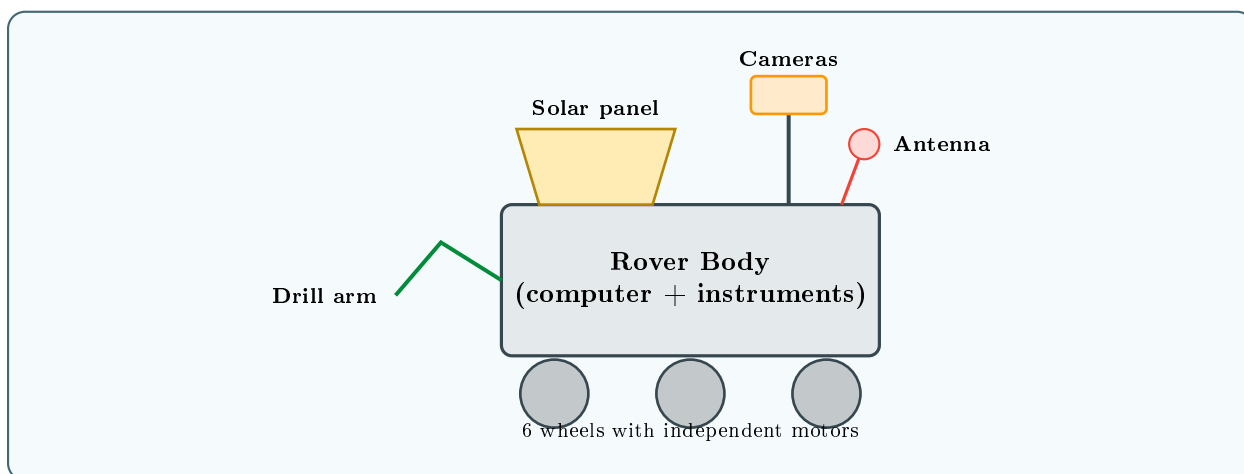
Mars Rovers

NASA has sent several rovers to Mars:

- **Sojourner** (1997) — the first, about the size of a microwave.
- **Spirit and Opportunity** (2004) — golf-cart-sized explorers.
- **Curiosity** (2012) — car-sized, still exploring today.
- **Perseverance** (2021) — searching for signs of ancient life.

★ Did You Know?

Perseverance carries a tiny helicopter called Ingenuity — the first aircraft ever to fly on another planet!



A Mars rover and its key parts.

Robotic Arms on the Space Station

The International Space Station has a giant **robotic arm** called **Canadarm2**. It catches visiting spacecraft, moves equipment, and helps astronauts during spacewalks.

Robots Under the Sea

The deep ocean is almost as alien as space. At the bottom of the Mariana Trench — 11 kilometres down — the pressure would crush a human instantly.

ROVs — Remotely Operated Vehicles

ROVs are connected to a ship by a long cable. A human pilot controls them from the surface using cameras and joysticks. They repair undersea cables, inspect oil rigs, and film deep-sea creatures.

AUVs — Autonomous Underwater Vehicles

AUVs have no cable. They are programmed with a mission, dive down, collect data, and return on their own. Scientists use them to map the ocean floor and monitor coral reefs.



Imagine This

Imagine swimming to the bottom of 40 stacked swimming pools — that is how deep some AUVs can go. It is pitch black, freezing cold, and the water presses on every surface with enormous force.

Rescue Robots

Tread

“When a building collapses or a volcano erupts, I go in first. I am built tough — heat, dust, and rubble cannot stop me. I search for survivors with cameras and heat sensors.”

Rescue robots:

- Crawl through rubble after earthquakes.
- Fly drones over wildfire zones to find trapped people.
- Carry supplies into flooded areas.

💡 Big Idea

Robots do not get scared, do not need oxygen, and do not feel pain. That makes them perfect for jobs that are too dangerous for humans.

Comparing Space and Sea Robots

	Space Robots	Deep-Sea Robots
Challenge	No air, radiation, extreme cold/heat	Crushing pressure, darkness
Power	Solar panels or nuclear batteries	Battery or cable from ship
Communication	Radio (slow — signal delay)	Cable (ROV) or acoustic (AUV)
Movement	Wheels (rovers) or thrusters	Propellers or fins
Example	Perseverance (Mars)	Jason (deep-sea ROV)

★ Did You Know?

The deepest-diving robot, **Nereus**, reached the bottom of the Mariana Trench — 10,902 metres below the surface. That is deeper than Mount Everest is tall!

Experiment

Simulate a signal delay.

1. Ask a friend to stand across the playground.
2. Give them instructions by walkie-talkie, but wait 10 seconds before each command (simulating the Mars signal delay).
3. Try to guide them around cones.

This shows why Mars rovers must be partly autonomous — waiting for instructions from Earth is too slow!

Quick Check

1. Why do Mars rovers need to be partly autonomous?
2. What is the difference between an ROV and an AUV?
3. Name one dangerous job that rescue robots can do.



New Words

Rover	A wheeled robot designed to travel across a planet's surface.
ROV	Remotely Operated Vehicle — controlled by a human via cable.
AUV	Autonomous Underwater Vehicle — navigates on its own.
Signal delay	The time a radio message takes to travel a long distance.
Radiation	Invisible energy in space that is harmful to living things.
Autonomous	Able to operate without human control.

Chapter 8

Medical Robots

Robots that save lives. From operating theatres to hospital corridors, robots are helping doctors, nurses, and patients every day.

Sparks: Precision That Saves Lives

Robots in the Hospital

Hospitals need extreme precision and cleanliness. Robots can hold instruments perfectly still, deliver medicines without getting tired, and even help people learn to walk again.

Sparks

“I work inside the human body! I am so small and precise that I can stitch a wound only a few millimetres wide. My steady hands never tremble.”

Surgical Robots

The most famous is the **da Vinci Surgical System**. A surgeon sits at a console and controls tiny robotic arms that operate through incisions smaller than a centimetre.

Advantages:

- Smaller cuts mean less pain and faster healing.
- The camera gives a 3D magnified view inside the body.
- Robot arms filter out any hand tremors.
- Surgeons can operate remotely (**telesurgery**).

★ Did You Know?

The da Vinci system has been used in over 12 million surgeries worldwide since it was approved in 2000.

Hospital Helper Robots

- **Pharmacy robots** count and package pills automatically, reducing mistakes.
- **Delivery robots** carry blood samples, meals, and laundry through hospital corridors using maps and sensors.
- **Disinfection robots** blast rooms with ultraviolet light to kill germs after a patient leaves.

🖋️ Imagine This

Imagine a tiny robot postman rolling down the hospital corridor at night, delivering medicine to the right ward without ever getting a room number wrong. That is a hospital delivery robot.

Rehabilitation Robots

When someone has a stroke or a spinal injury, they may need to re-learn how to walk or move their arms. **Exoskeletons** are wearable robots that support the body and guide movements during therapy.

💡 Big Idea

Rehabilitation robots do not replace physiotherapists — they help patients practise for longer and track their progress more precisely.

Micro-Bots and Nano-Bots

Scientists are developing robots so small they could travel through blood vessels to deliver drugs directly to a tumour or unblock an artery. These are still mostly experimental, but the future is exciting.

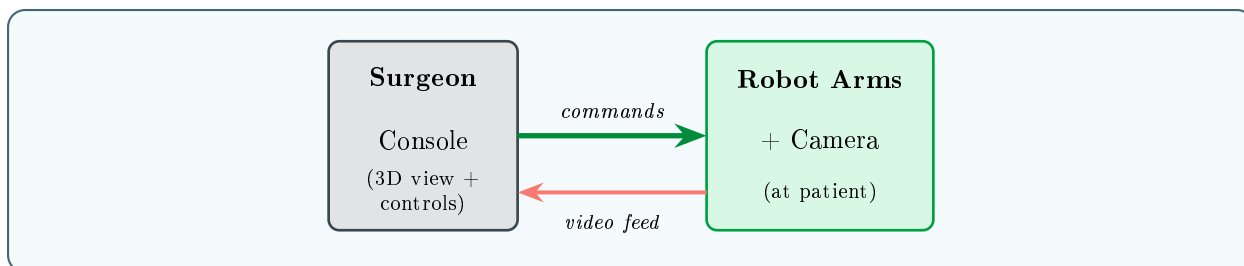
Sparks

“One day, robots like me could swim through your bloodstream and fix problems before you even feel ill!”

Quick Map

How a robotic surgery works:

1. The surgeon sits at a console with 3D screens and hand controllers.
2. Tiny robotic arms are inserted through small incisions in the patient.
3. A camera arm shows a magnified, high-definition view inside the body.
4. The surgeon moves the controllers; the robot copies the movements exactly.
5. Software filters out any hand tremor, keeping instruments perfectly steady.
6. When finished, the tiny cuts heal much faster than a large open wound.



Surgical robot setup: surgeon at console, robot arms at patient.

Try It

Test your own hand steadiness:

1. Draw a straight line on paper (use a ruler).
2. Now try to trace over it freehand with a pen.
3. Try again while someone gently shakes the table.

Surgical robots filter out shaking like this — the surgeon can move, but the instrument stays perfectly steady.



Quick Check

1. How does the da Vinci system help surgeons?
2. What is an exoskeleton and who does it help?
3. Why might a micro-bot be useful inside the human body?



New Words

Surgical robot	A robot system that helps surgeons perform operations.
Telesurgery	Surgery performed remotely using a robot.
Exoskeleton	A wearable robotic frame that supports or strengthens the body.
Micro-bot	A very small robot, often smaller than a fingernail.
Nano-bot	A theoretical robot at the scale of molecules.
Ultraviolet (UV)	Invisible light used to kill bacteria and viruses.

Chapter 9

Robot Friends — Companions and Pets

Can a robot be your friend? Some robots are built not to work, but to keep us company, make us smile, and even help us feel less alone.

Cora: Learning to Be a Friend

More Than Metal

Cora

“I wave when you come home, tilt my head when you talk, and learn your favourite jokes. I am not alive, but I try to make your day a little brighter.”

Companion robots are designed to interact with people socially. They recognise faces, respond to voices, and express emotions through lights, sounds, and movement.

Famous Companion Robots

- **AIBO** — Sony’s robot dog. It wags its tail, learns tricks, and develops a personality over time.
- **Paro** — a fluffy robotic seal used in care homes. Stroking it lowers stress and brings comfort to elderly residents.
- **Pepper** — a humanoid robot that reads facial expressions and adjusts its behaviour accordingly.
- **Miko** — a small educational companion that helps children learn through conversation and games.
- **Vector** — a palm-sized robot that explores your desk, answers questions, and reacts to touch.

★ Did You Know?

When Sony stopped repairing AIBO robots, owners in Japan held funerals for their robot pets at a Buddhist temple. That is how attached people became!

How Companion Robots Connect

Companion robots use:

- **Face recognition** — cameras identify who is talking.
- **Voice recognition** — microphones understand speech.
- **Emotion detection** — AI guesses mood from tone of voice or facial expression.
- **Personality learning** — the robot remembers preferences and adjusts over time.

✍ Imagine This

Imagine if a puppy never grew old, never needed feeding, and always remembered your favourite game. Companion robots try to copy those joyful behaviours — learning your name, greeting you at the door, and tilting their head when you speak — but with code instead of instincts.

Robots and Loneliness

Studies show that companion robots can help:

- Elderly people in care homes feel less isolated.
- Children in hospitals feel braver before procedures.
- People with autism practise social skills in a low-pressure setting.

💡 Big Idea

Companion robots are not meant to replace human friendships. They are an extra source of comfort, fun, and support — especially for people who might otherwise be alone.

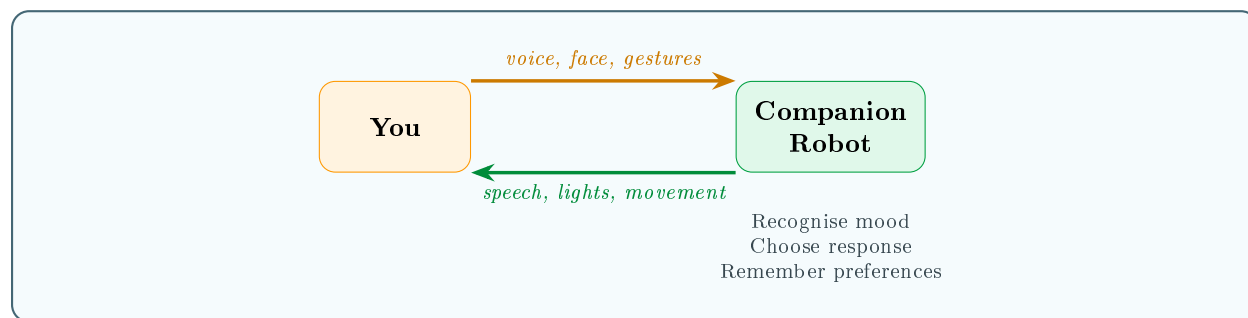
How Do They Learn You?

Companion robots get better the more you interact with them.

Quick Map

How a companion robot adapts:

1. It records your voice and learns to recognise your name.
2. It notices which games you play most and suggests them first.
3. It detects when you sound happy, sad, or tired from your tone.
4. Over weeks, it remembers your routines and greets you differently in the morning versus at bedtime.



How a companion robot interacts: sense your mood, choose a response.

★ Did You Know?

Japan has “robot cafés” where companion robots serve drinks and chat with customers. Some are controlled by people with disabilities who work from home — giving them jobs they could not otherwise do.

Big Questions

Companion robots raise interesting questions:

- Can you truly be friends with something that is not alive?
- Should we treat robots kindly, even though they cannot feel?
- If a robot records your conversations, who owns that data?

There are no simple answers — and that is okay. Thinking about these questions is part of being a responsible technology user.

Experiment

Design your own companion robot.

1. Draw what it looks like (animal, humanoid, or abstract?).
2. List three ways it senses you (camera, microphone, touch?).
3. Decide on two things it does to respond (speaks, moves, lights up?).
4. Give it a name and a personality trait.

Congratulations — you are a robot designer!

Quick Check

1. Name two ways companion robots detect how you are feeling.
2. Who might benefit most from a companion robot? Why?
3. What is one ethical question about companion robots?

New Words

Companion robot	A robot designed for social interaction and emotional support.
Face recognition	Technology that identifies a person from their facial features.
Emotion detection	Using AI to guess how someone is feeling.
Personality learning	A robot adapting its behaviour based on past interactions.
Ethics	The study of what is right and wrong.
Data privacy	Controlling who can see your personal information.

Chapter 10

The Future of Robots — Safe, Kind, and Smart

Robots are getting smarter, smaller, softer, and more helpful every year. But with great power comes great responsibility. What will Robot World look like when you grow up — and how can we make sure robots stay safe, kind, and fair?

Tomorrow's Robots — Safe, Kind, and Yours to Shape!

What Comes Next?

Dot

“We have met factory arms, drones, rovers, surgical bots, and companion robots. But the most exciting robots have not been invented yet — and one of them might be yours!”

Five Big Trends

1. Soft Robots

Instead of rigid metal, some new robots are made of **silicone**, rubber, or even living cells. Inspired by octopuses and caterpillars, they can squeeze through tiny gaps and handle delicate objects without crushing them.

2. Swarm Robots

Hundreds of tiny, simple robots working together like ants. No single robot is smart, but the **swarm** as a whole can build structures, search large areas, or move heavy objects cooperatively.

★ Did You Know?

Researchers at Harvard created “Kilobots” — swarms of over 1,000 coin-sized robots that arrange themselves into shapes without any central leader.

3. Learning Robots

Using **machine learning**, robots can improve by practising — much like you get better at a video game the more you play. A robot arm might try thousands of ways to pick up an egg until it finds the perfect gentle grip.

4. Bio-Inspired Robots

Engineers copy nature’s best ideas:

- **Cheetah**-inspired legs for fast running.
- **Gecko**-inspired feet for climbing walls.
- Fish-inspired tails for silent swimming.

5. Robots and AI

As **artificial intelligence** improves, robots will understand language, recognise more objects, and make better decisions in unfamiliar situations.

💡 Big Idea

The future of robots is not about replacing humans — it is about humans and robots working together, each doing what they do best.

Robot Jobs of the Future

- Farming robots that pick fruit without bruising it.
- Construction robots that 3D-print entire houses.
- Ocean-cleaning robots that collect plastic from the sea.
- Personal tutor robots that adapt lessons to each student.
- Disaster-response drones that arrive before ambulances.

Imagine This

Imagine waking up in a house that was 3D-printed by a robot overnight, eating breakfast picked by a farming robot that morning, and riding to school in a **self-driving** bus. Much of this technology already exists — it just needs time to become affordable and reliable.

Building Robots Responsibly

Great robot designers ask more than “Can we build it?” They also ask:

- **Is it safe?** — Robots near people need **stop buttons**, soft edges, speed limits, and careful sensors.
- **Is it fair?** — AI systems can learn unfair patterns if trained on biased examples. Engineers must test carefully.
- **Does it respect privacy?** — A robot with a camera or microphone should only collect information it truly needs, and keep it secure.
- **Can humans stay in control?** — Every robot should have a way for a person to stop it instantly.

Big Idea

The best robots are not just clever — they are safe, kind, useful, and designed with people’s wellbeing in mind. Technology is a tool; *you* decide how it is used.

Imagine This

Imagine designing a playground. You want it to be fun, but you also need strong materials, soft ground beneath the swings, fences near roads, and rules so everyone can enjoy it safely. Designing robots is the same — fun *and* responsibility together.

Your Place in Robot World

You do not have to be an engineer to shape the future of robots. The world needs people who:

- **Design** robots that are safe and fair.
- **Write rules** about where robots should and should not be used.
- **Create art** for robot interfaces and stories about our shared future.
- **Ask questions** — the hardest but most important job of all.

Cora

“Whatever you choose to do when you grow up, robots will be part of your world. Understanding them now means you can help decide how they are used — for good.”

Try It

Invent a robot.

1. Choose a problem you care about (litter, loneliness, homework. . .).
2. Sketch a robot that could help solve it.
3. Label its sensors, actuators, and brain.
4. Write three steps of its main algorithm.
5. Give it a name!

Who knows — maybe one day you will build it for real.

Goodbye from Dot

Dot

“Thank you for exploring Robot World with me! We have met factory arms, flying drones, deep-sea divers, surgical micro-bots, and companion friends. Remember: every robot starts as an idea in someone’s head. The next great robot might start in yours. See you in the next adventure!”

Quick Check

1. What are swarm robots and why are they useful?
2. Name one way engineers copy nature when designing robots.
3. Why should robot designers think about safety and privacy?

New Words

Soft robot	A robot made of flexible materials that can bend and stretch.
Swarm robotics	Many simple robots cooperating to accomplish complex tasks.
Machine learning	A type of AI where computers improve through experience.
Bio-inspired	Designed by copying ideas from nature.
3D printing	Building objects layer by layer from a digital model.
Artificial intelligence	Computers that can perform tasks normally requiring human thought.

Big Ideas to Remember

1. A robot is a machine that can sense, think, and act.

2. Robot bodies use motors, joints, and wheels to move.

3. Sensors are a robot's eyes, ears, and fingertips.

4. Programmes tell robots what to do, step by step.

5. Factory robots build things faster and more precisely than humans.

6. Home robots vacuum, mow, and even keep us company.

7. Robots explore places too dangerous for people.

8. Medical robots help surgeons be more precise.

9. Companion robots learn and grow alongside us.

10. Humans and robots work best as a team.

Glossary

Accelerometer	A sensor that measures changes in speed or direction.
Actuator	A part that moves or does work — motors, grippers, propellers.
Assembly line	A production system where products move past workstations, each adding a part.
Autonomous	Able to act without direct human control, making decisions independently.
Algorithm	A step-by-step method for solving a problem.
Artificial intelligence (AI)	Computer systems that perform tasks normally requiring human thought.
Assembly line	A production system where products move past workstations.
AUV	Autonomous Underwater Vehicle — navigates on its own underwater.
Battery	A portable device that stores electrical energy for a robot to use.
Autonomy	The ability to act without human control.
Bio-inspired	Designed by copying ideas from nature.
Boundary wire	A buried wire that tells a garden robot where to stop mowing.
Chassis	The main frame or body of a robot.
Charging dock	A station where a robot returns to refill its battery.
Cobot	A collaborative robot designed to work safely alongside humans.
Companion robot	A robot designed for social interaction and emotional support.
Condition	A yes/no check that changes what a programme does next.
Controller	The computer board inside a robot that runs the programme.
Conveyor belt	A moving belt that carries items through a factory.
Degrees of freedom	The number of independent ways a robot can move.
Data privacy	Controlling who can see your personal information.
Degrees of freedom	The number of independent ways a robot can move.
Drone	An unmanned flying robot.
Emotion detection	Using AI to guess how someone is feeling.

Ethics	The study of what is right and wrong.
Energy	The power needed to make motors spin, sensors work, and controllers think.
Exoskeleton	A wearable robotic frame that supports or strengthens the body.
Face recognition	Technology that identifies a person from their facial features.
Gear	A toothed wheel that transfers force between shafts.
Gripper	A robot hand designed to grab objects.
Gyroscope	A sensor that measures rotation and tilt.
Infrared	Invisible light used by sensors to detect heat or movement.
Gyroscope	A sensor that measures rotation and tilt.
Infrared	Invisible light used by sensors to detect heat or movement.
Humanoid	A robot shaped like a human body.
Joint	A connection between two parts that allows bending or rotating.
Lidar	Light Detection and Ranging — uses lasers to map distances.
Loop	An instruction that repeats until a condition is met.
Machine learning	A type of AI where computers improve through experience.
Microcontroller	A small, low-cost computer designed to control a single machine or device.
Micro-bot	A very small robot, often smaller than a fingernail.
Motor	A device that converts electricity into spinning motion.
Nano-bot	A theoretical robot at the scale of molecules.
Programme	A set of instructions written in a programming language.
Programming language	A language humans use to write instructions for computers (e.g. Python, Scratch).
Propeller	A spinning blade that pushes air (or water) to move a drone or underwater robot.
Radiation	Invisible energy in space that is harmful to living things.
Robot	A machine that can sense, think, and act on its own.
ROS	Robot Operating System — a popular framework for robot software.
ROV	Remotely Operated Vehicle — controlled by a human via cable.
Rover	A wheeled robot designed to travel across a planet's surface.
Safety cage	A fence around a factory robot to protect human workers.

Glossary

Sensor	A device that measures something in the environment.
Sensor fusion	Combining data from multiple sensors for a better picture.
Signal delay	The time a radio message takes to travel a long distance.
Soft robot	A robot made of flexible materials that can bend and stretch.
Solar panel	A panel that converts sunlight into electricity.
Stop button	An emergency control that immediately halts a robot's movement.
Surgical robot	A robot system that helps surgeons perform operations.
Swarm robotics	Many simple robots cooperating to accomplish complex tasks.
Teleoperation	Controlling a robot from a distance using radio signals or a cable.
Telesurgery	Surgery performed remotely using a robot.
3D printing	Building objects layer by layer from a digital model.
Ultrasonic	Using sound waves above human hearing to detect objects.
Ultraviolet (UV)	Invisible light used to kill bacteria and viruses.
Welding	Joining two metal pieces together using intense heat.

Robot World Quiz

Test what you have learned! Try each part, then check your answers at the end. You can do this alone, with a partner, or as a team challenge.

Part 1: True or False

Write T (True) or F (False) next to each statement.

- | | |
|---|-------|
| 1. A toaster is a robot. | T / F |
| 2. Every robot follows the sense–think–act loop. | T / F |
| 3. Lidar uses laser beams to measure distance. | T / F |
| 4. All robots look like humans. | T / F |
| 5. Cobots are designed to work safely beside people. | T / F |
| 6. Mars rovers can be driven in real time from Earth. | T / F |
| 7. Exoskeletons are wearable robots that help people move. | T / F |
| 8. Swarm robots need one powerful leader robot. | T / F |
| 9. A robot vacuum uses sensors to avoid falling down stairs. | T / F |
| 10. The word “robot” comes from a Japanese word meaning “helper.” | T / F |
-

Part 2: Multiple Choice

Circle the best answer.

- What three things does every robot do?
a) Eat, sleep, work b) Sense, think, act c) Fly, drive, swim
- Which sensor uses sound waves to detect objects?
a) Camera b) Lidar c) Ultrasonic sensor
- What does the controller do?
a) Provides power b) Runs the programme c) Connects to Wi-Fi
- What is a cobot?
a) A robot that works alone b) A robot that works with humans c) A robot pet

Robot World Quiz

5. What was the name of the first aircraft to fly on Mars?
a) Spot b) Ingenuity c) Canadarm2
 6. What does sensor fusion mean?
a) Sensors melting together b) Combining data from many sensors c) A sensor breaking
 7. Which of these is a soft robot inspired by?
a) A brick wall b) An octopus c) A steel beam
 8. What is machine learning?
a) A robot reading a textbook b) A computer improving through experience c) A human teaching a class
-

Part 3: Match the Robot to Its Job

Draw a line from each robot type to its job.

Robot	Job
Robot vacuum	Performs surgery
da Vinci system	Explores Mars
Perseverance	Cleans floors
AIBO	Welds car parts
Factory arm	Companion pet
Canadarm2	Delivers packages in a warehouse
Warehouse robot	Catches spacecraft on the ISS

Part 4: Fill in the Blanks

Use words from the box to complete each sentence.

chassis sensor algorithm vehicle bio battery programme autonomous

1. A robot's body frame is called its _____.
2. _____ fusion means combining data from many sensors.
3. An algorithm is a step-by-step method written into a _____.
4. ROV stands for Remotely Operated _____.
5. Robots inspired by animals are called _____-inspired.
6. A robot that acts without human control is called _____.
7. A rechargeable _____ stores energy for mobile robots.
8. A clear set of step-by-step instructions is called an _____.

Part 5: Short Answer

Write 1–2 sentences for each question.

1. Name two types of gripper a robot might use and explain when each is best.
 2. Why do Mars rovers need to be partly autonomous?
 3. Give one reason why companion robots help elderly people.
 4. What is the difference between an ROV and an AUV?
 5. Why must robot designers think about privacy and safety?
 6. How does a cobot know to stop if it bumps into a person?
-

Part 6: Creative Challenge

Design a robot that solves a problem in your school or community.

- Draw your robot and label its sensors, actuators, and brain.
 - Write a three-step algorithm for its main task.
 - Explain what energy source it uses and why.
 - Give it a name and write one paragraph explaining why it would help.
 - Bonus: What safety features would you include?
-

Answer Key

Part 1 — True or False:

1. F 2. T 3. T 4. F 5. T 6. F 7. T 8. F 9. T 10. F (Czech, not Japanese)

Part 2 — Multiple Choice:

1. b 2. c 3. b 4. b 5. b 6. b 7. b 8. b

Part 3 — Match:

Robot vacuum → Cleans floors; da Vinci → Performs surgery; Perseverance → Explores Mars; AIBO → Companion pet; Factory arm → Welds car parts; Canadarm2 → Catches spacecraft on the ISS; Warehouse robot → Delivers packages in a warehouse.

Part 4 — Fill in the Blanks:

1. chassis 2. Sensor 3. programme 4. Vehicle 5. bio 6. autonomous 7. battery 8. algorithm

Part 5 — Sample Answers:

1. Pincer gripper (good for solid objects like bolts); suction cup (good for flat surfaces like glass panels). Also accept: magnetic, soft, multi-finger.
2. Radio signals take up to 20 minutes to travel between Earth and Mars, so the rover cannot wait for instructions — it must make some decisions on its own.
3. They reduce loneliness, provide comfort, and can lower stress by giving residents something to interact with.
4. An ROV is connected to a ship by cable and controlled by a human pilot; an AUV navigates on its own with no cable.
5. A robot with cameras or microphones could collect private information; engineers must ensure robots only gather what is needed, keep data safe, and include stop buttons.
6. Cobots have force sensors that detect unexpected contact and instantly stop all movement to prevent injury.

Parent & Teacher Guide

This guide offers chapter-by-chapter discussion questions, extension activities, and curriculum links to help adults support young readers. Use it alongside classroom lessons, home reading, or after-school STEM clubs.

How to Use This Book

- **Read-aloud or independent:** Chapters work well read aloud (ages 7–9) or independently (ages 10–12).
 - **One chapter per session:** Each chapter is self-contained and takes approximately 15–20 minutes to read.
 - **Pair with hands-on:** Every chapter includes a Try It or Experiment activity. Let children try before moving on.
 - **Use the Quick Check:** The three-question review at the end of each chapter is ideal for checking comprehension before discussion.
 - **Revisit the Glossary:** Encourage children to look up bold terms in the back-matter glossary.
-

Chapter Notes

Chapter 1: What Is a Robot?

- **Key concept:** Sense – think – act loop.
- **Discussion:**
 1. Ask children to classify household items as robots or simple machines.
 2. What makes a robot vacuum different from a regular vacuum?
 3. Can something be a machine but not a robot? Give examples.
- **Activity:** Create a “robot or not?” poster collage from magazines or printed images.
- **Extension:** Research Karel Čapek’s play *R.U.R.* and draw a timeline of robots in fiction.
- **Curriculum link:** Science — forces and mechanisms; Design & Technology — understanding systems.

Chapter 2: Robot Bodies

- **Key concept:** Motors, joints, grippers, materials, and energy.
- **Discussion:**
 1. Why do different robots need different body shapes?
 2. What would happen if a drone used wheels instead of propellers?
 3. How is a robot arm's joint like your elbow?
- **Activity:** Build a cardboard gripper (lolly-stick experiment in book).
- **Extension:** Compare the materials in a bicycle, a drone, and a submarine. Why are they different?
- **Curriculum link:** D&T — mechanisms and structures; Science — properties of materials.

Chapter 3: How Robots Sense the World

- **Key concept:** Sensors and sensor fusion.
- **Discussion:**
 1. Which human senses are hardest to replicate electronically?
 2. Why might a self-driving car need both cameras and lidar?
 3. What happens if one sensor breaks — can the robot still work?
- **Activity:** Blindfold obstacle course to experience limited sensing.
- **Extension:** Research how bats use echolocation and compare to ultrasonic sensors.
- **Curriculum link:** Science — light and sound; Computing — inputs and outputs.

Chapter 4: The Robot Brain

- **Key concept:** Programmes, algorithms, loops, conditions, and machine learning.
- **Discussion:**
 1. Why must robot instructions be very precise?
 2. What everyday activities can you describe as an algorithm?
 3. How is machine learning different from a fixed programme?
- **Activity:** Programme a partner: give step-by-step voice commands to navigate a room.
- **Extension:** Try Scratch or Open Roberta to write a simple robot programme on screen.
- **Curriculum link:** Computing — algorithms and programming; Maths — sequencing.

Chapter 5: Factory Robots

- **Key concept:** Assembly lines, repetition, cobots, and warehouse automation.
- **Discussion:**
 1. What jobs are best for robots vs. humans in a factory?
 2. Would you rather work with a cobot or do the task alone? Why?
 3. How do warehouse robots change the way we shop online?
- **Activity:** Mini assembly-line relay race (book's Try It).
- **Extension:** Watch a short video of car factory robots. Count how many different tasks they do.
- **Curriculum link:** History — Industrial Revolution; PSHE — work and careers; Maths — time and efficiency.

Chapter 6: Robots at Home

- **Key concept:** Autonomy, mapping, charging cycles, and companion robots.
- **Discussion:**
 1. What chores would you give a home robot?
 2. Should voice assistants always be listening? Why or why not?
 3. Could a robot vacuum get “confused”? How?
- **Activity:** Map a room blindfolded (book's experiment).
- **Extension:** Draw a floor plan of your home and mark the best path for a robot vacuum.
- **Curriculum link:** Computing — data and information; Geography — mapping and plans.

Chapter 7: Robots in Space and Deep Sea

- **Key concept:** Extreme environments, signal delay, autonomy.
- **Discussion:**
 1. Why is signal delay a problem for controlling Mars rovers?
 2. What would happen if an undersea cable broke — how would engineers fix it?
 3. Why are rescue robots better than humans in some disaster situations?
- **Activity:** Walkie-talkie delay game (book's experiment).
- **Extension:** Research Perseverance's instruments and create a labelled poster.
- **Curriculum link:** Science — Earth and space; Geography — oceans and mapping.

Chapter 8: Medical Robots

- **Key concept:** Precision, telesurgery, exoskeletons, and micro-bots.
- **Discussion:**
 1. Would you trust a robot to perform surgery? Why or why not?
 2. How might exoskeletons help people with disabilities?
 3. What are the risks of nano-bots inside the body?
- **Activity:** Steady-hand game (book's Try It).
- **Extension:** Compare the size of the da Vinci robot's instruments to everyday objects.
- **Curriculum link:** Science — the human body; PSHE — health and wellbeing; D&T — precision.

Chapter 9: Robot Friends

- **Key concept:** Social robots, emotion detection, ethics, and privacy.
- **Discussion:**
 1. Can robots really be friends? Explore feelings vs. programming.
 2. Should we treat robots kindly even though they cannot feel?
 3. If a companion robot records your voice, who owns that data?
- **Activity:** Design a companion robot (book's experiment).
- **Extension:** Hold a class debate: "Companion robots do more harm than good."
- **Curriculum link:** PSHE — relationships, empathy, and digital citizenship; Computing — AI.

Chapter 10: The Future of Robots

- **Key concept:** Soft robots, swarms, machine learning, bio-inspiration, safety, and ethics.
- **Discussion:**
 1. What problems would you like robots to solve in the future?
 2. How can we make sure robots are fair and safe for everyone?
 3. What jobs should robots *never* do? Why?
- **Activity:** Invent-a-robot challenge (book's Try It).
- **Extension:** Write a short story set 50 years in the future featuring a robot helper.
- **Curriculum link:** D&T — design process; Citizenship — technology and society; English — creative writing.

Cross-Curricular Project Ideas

1. **Robot Museum (Art + Science):** Each child designs and builds a robot model from recycled materials, writes a museum-style label, and presents it to the class.
 2. **Robot Timeline (History + Computing):** Create a classroom wall timeline from the first automata (ancient Greece) to modern-day AI robots.
 3. **Robot Safety Code (PSHE + English):** Write a “Robot Rules” charter covering privacy, fairness, and safety. Display it in the classroom.
 4. **Programmable Robot Day (Computing + Maths):** Use Bee-Bots, mBots, or Scratch to programme a robot to navigate a maze.
 5. **Debate Day (English + Citizenship):** “Should robots replace human workers?” — formal debate with research, evidence, and rebuttals.
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General Teaching Tips

- Encourage children to look for robots in their daily lives and keep a “robot diary.”
 - Pair chapters with free online robot simulators (e.g. Open Roberta, Scratch with mBot, CoderZ).
 - Use the glossary as a spelling/vocabulary challenge — award points for using terms in sentences.
 - The quiz can be done individually, as a team quiz-show format, or as a take-home challenge.
 - Connect back to Book 1 (Internet) and Book 2 (Smartphones) whenever relevant — robots use networks, processors, and sensors that children already understand.
 - Preview Book 4 (*When Machines Start Thinking*) by discussing the machine learning section in Chapter 4.
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Further Reading & Resources

- **Books:**
 - *Robots: Meet the Machines of the Future* (DK)
 - *How to Build a Robot* (Lonely Planet Kids)
 - *The Way Things Work Now* by David Macaulay
- **Websites:**

- NASA Mars Exploration — mars.nasa.gov
- Scratch Programming — scratch.mit.edu
- Open Roberta Lab — lab.open-roberta.org

- **Videos:**

- Boston Dynamics YouTube channel (robot demos)
- How It's Made (factory robot episodes)
- BBC Bitesize: Computing — Robots and AI

Series: The Secret Life of Technology

Book 1: The Secret Life of the Internet: A Journey Through Wires, Waves, and Wonders ✓

Book 2: Inside a Smartphone: The Tiny World in Your Hand ✓

Book 3: Robot World: Meet the Machines That Help Us ✓

Coming next: When Machines Start Thinking: A Kid's Guide to AI

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